

MRINAL GUPTA

Computer Science and Engineering

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Education

Degree	Institute	Year	CGPA/%
M.Tech (Cloud Computing)	Indian Institute of Technology, Patna	2026 - Present	
Executive PG (AI & ML)	Indian Institute of Technology, Kharagpur	2026 - 2027	
B.Tech (Computer Engineering)	MIT Academy of Engineering, Pune	2021 - 2025	9.56
Intermediate	D.A.V Public School, New Panvel	2020	84%
Matriculation	New Horizon Public School, New Panvel	2018	88.8%

Professional Summary

Full-stack software engineer with hands-on experience building and deploying production-grade systems using ReactJS, Spring Boot, MySQL, and AWS. Proficient in REST API design, CI/CD automation, and cloud-native architecture. GATE 2026 — 98th percentile. Pursuing M.Tech (Cloud Computing) at IIT Patna and Executive PG (Applied AI & ML) at IIT Kharagpur. Seeking SDE / backend engineering roles where I can contribute from day one.

Publications

A Comparative Study of Deep Learning Pre-Trained Models: **ResNet50**, **EfficientNetB7**, **DenseNet-121** for Remote Sensing Image Classification is presented, and under review for publication.

Work Experiences

Uniconverge Technologies Pvt. Ltd.

[May 2024 – Jun 2024]

Cloud Computing Internship | Remote | Summer Internship 2024

- Architected and deployed a full-stack healthcare portal on AWS, automating the release pipeline using **CodePipeline**, **CodeBuild**, and **CodeDeploy**. Configured Application **Load Balancer** and **Auto Scaling Groups** for high-availability
- Managed static assets via **S3**.

MIT Academy of Engineering

[Jul 2023 – Aug 2023]

Web Development Internship | Remote | Summer Internship 2023

- Engineered a web-based visitor pass management system using **Spring MVC** and **MySQL**, implementing role-based access for security staff and administrators, reducing manual entry overhead for 50+ daily campus visitors.

Teachnook (with IIT Bhubaneswar)

[Aug 2022 – Sep 2022]

Web Development Internship | Remote | Summer Internship 2022

- Developed 5+ responsive web pages using **HTML**, **CSS**, and **JavaScript** as part of a structured full-stack curriculum in collaboration with IIT Bhubaneswar, strengthening foundations in **DOM manipulation** and **UI/UX** principles.

Projects

DetectiveDB

[Feb 2025 – May 2025]

Guide: Dr. M. N. Vharkate | Capstone Project

- Built a case-based learning platform for DBMS concepts using **ReactJS**, **Spring Boot**, and **MySQL**, deployed on AWS with infrastructure managed via Terraform.
- Designed 8+ normalized relational schemas with indexed queries, achieving sub-100ms retrieval for case scenario lookups.
- Implemented RESTful APIs for scenario navigation and user progress tracking across 20+ case modules.

Surya

[Sep 2024 – Nov 2024]

Guide: Ms. S. B. Kharat | OpenCV Project

- Developed a real-time computer vision system to detect and correct Surya Namaskar yoga poses using **OpenCV**, **Flask**, and **ReactJS**.
- Achieved $\approx 85\%$ pose classification accuracy by computing joint angles across 12 body keypoints in real time using OpenCV.
- Delivered corrective audio-visual feedback within 200ms latency, enabling live session use.

Suvidha Healthcare Management System

[May 2024 – Jun 2024]

Cloud Computing Internship Project | Summer Internship 2024

- Developed a full-stack healthcare portal using **Spring MVC, ReactJS, and MySQL** with role-based access for patients, doctors, and admins.
- Reduced deployment time from manual hours to under 10 minutes via fully automated **AWS CI/CD pipeline** with zero-downtime rolling updates.
- Deployed on AWS with S3 static hosting, **Application Load Balancer, and Auto Scaling Groups** for production-grade availability.

Chemical Industry Management System

[Mar 2024 – Apr 2024]

Guide: Dr Manish Giri | Collaboration with Seemans Pumps and Equipments Pvt. Ltd.

- Built a production and inventory management platform using **Spring Boot, MySQL, and ReactJS**.
- Exposed 10+ REST endpoints handling full order lifecycle — creation, inventory deduction, and fulfillment tracking.
- Handled **CORS configuration** and integrated frontend-backend communication for seamless multi-role workflows.

Projekta - Project Management Tool

[Sep 2022 – June 2023]

Guide: Dr. C. H. Nemade | Minor Project

- Developed a collaborative project management tool for task assignment, progress tracking, and team coordination using **Spring MVC and MySQL**.
- Designed relational schema to support multi-user project hierarchies with role-based task visibility.

Technical Skills

Programming Languages	Java, C++, C, Python
Web Technologies	ReactJS, Tailwind CSS, HTML, CSS, Flask, OpenCV
Tools	Git, GitHub, Linux, OOPs, Postman, IntelliJ IDE
Databases	MySQL, NoSQL, DynamoDB, Amazon RDS
Backend & Frameworks	Spring Framework, Spring Boot, Spring MVC, JDBC, REST APIs, Microservices, Distributed Systems
Cloud & DevOps	AWS, CI/CD, Jenkins, Docker, CodePipeline, Terraform, Infrastructure as Code (IaC), Load Balancing
CS Fundamentals	Data Structures, Algorithms, DBMS, Computer Networks, Agile/Scrum

Certifications & Achievements

AWS Academy Graduate – Microservices and CI/CD Pipeline Builder	[2024]
AWS Academy Graduate – Data Engineering	[2024]
Red Hat Academy – Program Learner	[2024]
Postman API Fundamentals Student Expert	[2024]
Oracle Certified Associate – OCI Foundations	[2023]
Microsoft Azure Fundamentals (AZ-900)	[2023]

Best Outgoing Student 2024–25

Qualified GATE 2026 – Percentile: 98.0822

Qualified GATE 2025 – Score: 635

Qualified GATE 2024 – Score: 367

Top 5% in NPTEL

Social Links

- **LinkedIn:** [linkedin.com/in/mrinal03](https://www.linkedin.com/in/mrinal03)
- **GitHub:** github.com/mrinal4703
- **Personal:** mrinal03.vercel.app
- **Location:** Pune District, Maharashtra, India